A Comprehensive Literature Survey on Zero-Shot Parkinson's Disease Detection from Speech

Comparing Large Language and Audio Models

Systematic Literature Survey

Research Focus: Zero-Shot Speech-Based PD Detection
Input Modalities: Acoustic Features vs. Raw Audio
Model Classes: LLMs, LALMs and LARMs
Languages: Bengali, English, Italian and Spanish

Literature Survey Report

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1 Introduction and Research Context

1.1 Background

The rapid emergence of large language models (LLMs) and large audio language models (LALMs) has opened new frontiers in clinical speech processing. These foundation models are trained on vast quantities of data using self-supervised or weakly supervised learning and have demonstrated remarkable zero-shot reasoning capabilities across a wide range of domains, including healthcare-related applications.

In the context of Parkinson's Disease (PD) detection, speech analysis has emerged as one of the most actively investigated digital biomarker paradigms. Speech impairments associated with PD are commonly described as hypokinetic dysarthria and can affect a large proportion of individuals with the disease. These impairments may also appear before prominent motor symptoms, making speech a potentially useful non-invasive biomarker.

Common vocal and speech manifestations include:

- **Reduced vocal intensity (hypophonia):** Reduced loudness of speech.
- **Monotonic prosody:** Reduced variation in pitch and intonation.
- **Articulatory imprecision:** Reduced precision in the production of speech sounds.
- **Increased jitter and shimmer:** Increased cycle-to-cycle variation in frequency and amplitude.
- **Altered harmonic-to-noise ratio:** Changes in the balance between periodic and noise components of the voice.
- **Disrupted temporal patterns:** Changes in speech rate, pauses and rhythmic organization.

These characteristics provide a physiological basis for investigating speech as a digital biomarker for PD.

1.2 The Promise of Zero-Shot Learning

Traditional approaches to speech-based PD detection have primarily relied on supervised machine learning. These approaches generally fall into two broad categories: handcrafted acoustic features combined with classical classifiers, and end-to-end deep learning models.

Such approaches typically require:

- **Large labeled datasets** for individual languages and recording conditions.
- **Task-specific fine-tuning** of machine learning or deep learning models.
- **Careful dataset-specific optimization** of features, hyperparameters and decision thresholds.

Zero-shot approaches based on foundation models provide an alternative paradigm. Instead of training a model specifically for PD classification, a pretrained model is given an instruction and asked to make a prediction without task-specific training.

Potential advantages include:

1. **Rapid deployment:** A single pretrained model can potentially be evaluated across multiple languages and recording conditions.
2. **Reduced labeled-data requirements:** No task-specific training dataset is required during inference.
3. **Potential cross-lingual generalization:** Large pretrained models may contain representations learned from diverse languages.
4. **Flexible deployment:** Prompt-based inference can be adapted to different classification tasks without retraining the complete model.

However, zero-shot capability does not necessarily imply robust clinical performance. The central question is whether representations learned from broad speech data contain sufficient information about pathological speech characteristics to support reliable PD detection.

1.3 Key Research Question

Despite the rapid development of large language and audio models, an important question remains:

> **How does the input modality—handcrafted acoustic features versus raw audio waveform—influence zero-shot Parkinson's disease detection performance across different languages?**

This question is important because input representation determines:

- which model architectures can be used;

- how acoustic and clinical information is represented;
- the interpretability of model decisions;
- computational requirements;
- sensitivity to language and recording conditions; and
- the ability of models to capture temporal and spectral information.

2 The Problem: Modality Effects in Zero-Shot PD Detection

2.1 Two Competing Paradigms

The emergence of large language models has resulted in two distinct approaches to zero-shot speech-based PD detection.

Table 1: Comparison of feature-based and audio-based zero-shot paradigms.

Paradigm	**Input**	**Model Type**	**Approach**
Feature-based	Handcrafted acoustic features serialized as text	General-purpose LLM	Provide structured feature-value pairs as textual input.
Audio-based	Raw audio waveform	LALM / LARM	Process the waveform directly using an audio-capable multimodal model.

The feature-based approach effectively converts an acoustic representation into a textual representation that can be processed by a conventional LLM. The audio-based approach instead provides the raw waveform directly to a model capable of processing audio.

2.2 Gap in the Literature

Prior to this study, there was limited systematic investigation comparing zero-shot inference across different speech representations under a unified experimental framework and across multiple languages.

Existing research has predominantly:

- relied on supervised machine learning for PD detection;
- investigated handcrafted acoustic features;
- evaluated deep learning models using task-specific training;
- investigated LLMs primarily for clinical care systems and decision support;
- investigated audio language models across general audio tasks; and
- evaluated individual foundation models without directly comparing input modalities.

Consequently, it remained unclear whether a zero-shot general-purpose LLM supplied with carefully designed acoustic features could compete with audio-capable foundation models receiving raw speech.

2.3 Clinical and Practical Implications

Understanding modality effects has several practical implications.

1. **Deployment Context:**

   Feature-based systems require an additional feature-extraction pipeline, whereas audio-based models can potentially process raw recordings directly.

2. **Interpretability:**

   Handcrafted acoustic features have known physiological and clinical interpretations. In contrast, latent representations learned directly from raw audio are generally more difficult to interpret.

3. **Language Adaptation:**

   Feature-based and audio-based systems may exhibit substantially different levels of cross-lingual robustness.

4. **Computational Resources:**

   Acoustic feature extraction introduces preprocessing requirements, whereas large audio models may require substantially greater computational resources during inference.

3 Methodology: A Unified Zero-Shot Evaluation Framework

3.1 Datasets: Four Languages and Diverse Speech Tasks

The study evaluated four Parkinson's disease speech datasets covering four languages and several different speech tasks.

Table 2: Datasets used in the unified zero-shot evaluation framework.

Dataset	**Language**	**PD**	**HC**	**Speech Task**	**Key Characteristics**
BenSParX	Bengali	60	60	Conversation	Low-resource language; spontaneous conversational speech.
MDVR-KCL	English	16	21	Reading text	Controlled recording conditions; read speech.
IPVS	Italian	28	22	Text-dependent utterance	Predefined repeated sentences.
NeuroVoz	Spanish	23	53	Spontaneous speech	Free speech; Castilian Spanish.

3.1.1 Dataset Selection Rationale

The four datasets were selected to provide variation along several dimensions.

Table 3: Rationale for selecting the four datasets.

Dimension	Rationale
Linguistic Diversity	Bengali, English, Italian and Spanish represent different linguistic families and different levels of representation in pretrained speech models.
Task Diversity	Conversation, reading, text-dependent utterances and spontaneous speech provide different levels of linguistic and acoustic complexity.
Sample Size	The datasets range from small to moderate sample sizes, approximately 37–120 speakers.
Recording Conditions	The datasets contain different recording protocols and environmental conditions.

The inclusion of Bengali is particularly important because it represents a relatively low-resource language compared with English and several major European languages that are more strongly represented in large-scale speech datasets.

3.2 Feature Extraction: Standardized Handcrafted Features

For the feature-based approach, the study extracted 71 handcrafted acoustic features using a standardized processing pipeline.

Table 4: Handcrafted acoustic feature categories.

Feature Category	Examples	Clinical Relevance
Time-domain	Jitter, shimmer and HNR	Perturbation measures sensitive to dysphonia and voice instability.
Frequency-domain	Spectral descriptors	Capture formant structure and harmonic characteristics.
Cepstral	MFCCs	Represent spectral and articulatory characteristics of speech.

3.2.1 Standardization Strategy

Several design decisions were used to ensure that differences between approaches could be interpreted primarily as modality effects.

- Non-overlapping 10-second speech segments were used.

- The same feature configuration was applied across all corpora.
- No dataset-specific feature engineering was performed.
- The same 71-dimensional representation was used for all feature-based evaluations.

This standardization is important because corpus-specific feature optimization could otherwise introduce another source of performance variation.

3.3 Models: Three Categories of Large-Scale Models

The study investigated three broad categories of large-scale models.

3.3.1 Category 1: Text-Based LLMs

The feature-based paradigm used a general-purpose large language model.

Table 5: Text-based LLM used for feature-based zero-shot classification.

Model	**Architecture**	**Parameters**	**Role**
LLaMA 3	Decoder-only Transformer	8 billion	Processes serialized acoustic feature-value pairs.

The 71-dimensional acoustic feature vector was converted into a structured textual representation. The LLM was then prompted to classify the subject as either a healthy control or an individual with PD.

- **Input:** 71-dimensional acoustic feature representation.
- **Representation:** Structured textual list of feature-value pairs.
- **Output:** Binary classification.
- **Class labels:** 0 = Healthy, 1 = Parkinson's disease.
- **Decoding:** Deterministic with temperature equal to zero.

3.3.2 Category 2: Large Audio Language Models

The second category consisted of large audio language models capable of processing raw waveform information.

Table 6: Large audio language models used in the study.

Model	Architecture	Key Feature
Qwen2-Audio	Audio encoder with Qwen language model	Speech-and-text-to-text multimodal processing.
Pengi	Pretrained audio encoder with frozen LLM	Maps continuous audio embeddings into a language-model space.

These models received raw audio together with a textual instruction describing the classification task.

3.3.3 Category 3: Large Audio Reasoning Models

The third category consisted of large audio reasoning models.

Table 7: Large audio reasoning model used in the study.

Model	Architecture	Key Feature
Audio-Reasoner	Cross-modal alignment and reasoning	Higher-level reasoning over acoustic information.

Audio-Reasoner receives a raw waveform together with a binary classification prompt and generates a prediction. Its distinguishing characteristic is the explicit emphasis on reasoning over audio information.

3.4 Prompt Construction

Prompt construction is a critical component of zero-shot learning because the model does not receive task-specific parameter updates.

3.4.1 Feature-Based Prompt

The LLaMA 3 prompt was structured around the clinical classification task.

```
Task: You are a clinical classification model. Based on the audio
features extracted from a person's speech, classify whether a person
has Parkinson's disease or not.
```

The prompt further instructed the model to output exactly one binary label.

```
Output 1 if the person has Parkinson's disease, or 0 if the person
is healthy. Respond with exactly one token: 0 or 1.
```

The feature vector was subsequently inserted into the prompt as a list of feature-value pairs.

3.4.2 Audio-Based Prompt

The audio-based models received a clinically informed instruction.

```
You are an audio analysis model. Your task is to decide whether
the speech characteristics are more consistent with healthy control
speech or Parkinson's. Consider acoustic cues, including pitch variability,
loudness variability over time, articulation precision of consonants,
voice quality, speech rate, and rhythm. Make a balanced decision
based only on the provided audio. Output only a single digit: 0
= Healthy or 1 = Parkinson's disease.
```

3.4.3 Prompt Design Principles

The prompts followed several principles:

1. Standardized instruction-style prompting.
2. Clinically informed acoustic cues for audio-based models.
3. Binary output constraints.
4. No additional patient-level clinical information.
5. No task-specific training.
6. Consistent classification objective across models.

These choices were intended to isolate the effect of input modality rather than providing one model with substantially more information than another.

3.5 Evaluation Protocol

3.5.1 Subject-Level Aggregation

The models operated on individual speech segments, but the final evaluation was performed at the subject level.

The procedure consisted of:

1. Segmenting recordings into non-overlapping 10-second segments.
2. Generating a prediction for each segment.
3. Aggregating segment-level predictions for each subject.
4. Applying majority voting to determine the subject-level label.
5. Using the label with the higher mean predicted probability to break ties.

This procedure avoids treating multiple segments from the same individual as independent subjects during final evaluation.

3.5.2 Evaluation Metrics

Table 8: Evaluation metrics used in the study.

Metric	**Purpose**
Balanced Accuracy	Accounts for class imbalance by averaging sensitivity and specificity.
AUROC	Measures discrimination across classification thresholds.
Sensitivity	Measures the proportion of PD subjects correctly detected.
Specificity	Measures the proportion of healthy controls correctly identified.
Brier Score	Measures the accuracy of probabilistic predictions and therefore provides information about calibration.

Balanced accuracy is particularly useful for these datasets because the number of PD and healthy participants is not always identical.

The Brier score is important from a clinical perspective because a model can discriminate between classes while still producing poorly calibrated probabilities.

3.5.3 Statistical Rigor

The study used bootstrap-based confidence intervals to quantify uncertainty.

Table 9: Statistical evaluation procedure.

Component	**Configuration**
Confidence Interval	95% confidence intervals.
Bootstrap Replicates	10,000 replicates.
Bootstrap Method	Bias-corrected and accelerated nonparametric bootstrap.
Class Preservation	Class proportions preserved during bootstrap sampling.
Random Seed	Fixed seed of 0.
Decoding	Deterministic, temperature = 0.
Evaluation Frequency	Each recording evaluated once per model.

The use of a large number of bootstrap replicates provides a more stable estimate of uncertainty, particularly for small datasets.

4 Results and Key Findings

4.1 BenSParX: Bengali Conversational Speech

BenSParX represents the low-resource language condition and consists of Bengali conversational speech.

Table 10: Zero-shot performance on the BenSParX Bengali dataset.

Model	Type	B. Accuracy	AUROC	Sensitivity	Specificity
LLaMA 3	Feature-based	**83.33%**	**0.901**	86.67%	80.00%
Qwen2-Audio	Audio-based	50.00%	0.536	16.67%	83.33%
Pengi	Audio-based	58.33%	0.617	45.00%	71.67%
Audio-Reasoner	Audio-based	50.83%	0.534	91.67%	10.00%

Model	Brier Score
LLaMA 3	**0.228**
Qwen2-Audio	0.258
Pengi	0.258
Audio-Reasoner	0.396

4.1.1 Key Findings

The feature-based LLaMA 3 approach dramatically outperformed the audio-based models on Bengali speech.

LLaMA 3 achieved an AUROC of 0.901, while the audio-based models remained close to chance-level discrimination, with AUROC values between 0.534 and 0.617.

Audio-Reasoner exhibited an extreme sensitivity-specificity trade-off. It achieved 91.67% sensitivity but only 10.00% specificity. Therefore, it identified most PD participants but incorrectly classified a very large proportion of healthy controls as PD.

The lower Brier score of LLaMA 3 also indicates better calibration compared with the audio-based models.

4.1.2 Interpretation

Several factors may contribute to the strong performance of the feature-based approach.

- Bengali is relatively underrepresented in many large-scale speech pretraining datasets.
- Handcrafted acoustic features may capture language-independent properties of pathological phonation.
- The feature representation removes some language-specific lexical information.

- Raw audio foundation models may therefore have difficulty transferring their learned representations to Bengali pathological speech.

Thus, the Bengali results provide evidence that handcrafted acoustic representations can remain highly competitive in low-resource linguistic settings.

4.2 MDVR-KCL: English Controlled Reading

MDVR-KCL provides a contrasting setting involving English read speech recorded under relatively controlled conditions.

Table 11: Zero-shot performance on the MDVR-KCL English dataset.

Model	Type	B. Accuracy	AUROC	Sensitivity	Specificity
LLaMA 3	Feature-based	50.74%	**0.702**	6.25%	95.24%
Qwen2-Audio	Audio-based	52.98%	0.506	25.00%	80.95%
Pengi	Audio-based	29.76%	0.268	50.00%	9.52%
Audio-Reasoner	Audio-based	**69.49%**	0.609	43.75%	**95.24%**

Model	Brier Score
LLaMA 3	0.242
Qwen2-Audio	0.248
Pengi	0.442
Audio-Reasoner	0.252

4.2.1 Key Findings

Audio-Reasoner achieved the highest balanced accuracy at 69.49%.

LLaMA 3 nevertheless achieved the highest AUROC at 0.702, indicating stronger threshold-independent discrimination despite its low sensitivity at the default decision threshold.

Pengi exhibited the weakest overall performance, with an AUROC of only 0.268 and a Brier score of 0.442.

The results demonstrate that balanced accuracy and AUROC can tell different stories. A model may possess useful ranking or discrimination ability while producing a poor classification threshold.

4.2.2 Interpretation

The stronger performance of Audio-Reasoner may reflect several characteristics of the English dataset:

- English speech is extensively represented in large-scale speech pretraining.
- Controlled reading provides relatively clean and consistent acoustic signals.
- Raw audio models can directly access temporal and spectral information.
- Reasoning-enhanced models may be able to combine multiple acoustic cues.

However, the relatively modest AUROC indicates that the model does not provide uniformly strong discrimination.

4.3 IPVS: Italian Text-Dependent Utterances

The IPVS dataset contains Italian speech involving predefined repeated utterances.

Table 12: Zero-shot performance on the IPVS Italian dataset.

Model	Type	B. Accuracy	AUROC	Sensitivity	Specificity
LLaMA 3	Feature-based	51.79%	**0.805**	3.57%	**100%**
Qwen2-Audio	Audio-based	**54.87%**	0.472	**64.29%**	45.45%
Pengi	Audio-based	32.63%	0.188	10.71%	54.55%
Audio-Reasoner	Audio-based	51.95%	0.395	85.71%	18.18%

Model	Brier Score
LLaMA 3	0.249
Qwen2-Audio	0.254
Pengi	0.444
Audio-Reasoner	0.398

4.3.1 Key Findings

LLaMA 3 obtained the highest AUROC at 0.805 but exhibited extreme specificity bias. Its sensitivity was only 3.57%, while specificity was 100%.

Qwen2-Audio achieved the highest balanced accuracy at 54.87%, but its AUROC was only 0.472.

Pengi again demonstrated poor performance, achieving an AUROC of only 0.188.

Audio-Reasoner exhibited the opposite behavior from LLaMA 3, with high sensitivity of 85.71% but very low specificity of 18.18%.

4.3.2 Interpretation

The Italian results demonstrate the difference between discrimination and threshold behavior.

A high AUROC indicates that a model can potentially distinguish between the classes when an appropriate threshold is selected. However, the default zero-shot decision threshold may be poorly aligned with the underlying score distribution.

Therefore:

> High AUROC does not necessarily imply high balanced accuracy under a fixed zero-shot decision rule.

This distinction is particularly important when zero-shot models are used without calibration or task-specific threshold optimization.

4.4 NeuroVoz: Spanish Spontaneous Speech

NeuroVoz represents spontaneous Castilian Spanish speech.

Table 13: Zero-shot performance on the NeuroVoz Spanish dataset.

Model	Type	B. Accuracy	AUROC	Sensitivity	Specificity
LLaMA 3	Feature-based	52.58%	0.486	39.13%	66.04%
Qwen2-Audio	Audio-based	63.04%	0.519	26.09%	**100%**
Pengi	Audio-based	53.04%	0.517	17.39%	88.68%
Audio-Reasoner	Audio-based	**63.90%**	**0.676**	**39.13%**	88.68%

Model	Brier Score
LLaMA 3	0.247
Qwen2-Audio	0.219
Pengi	0.235
Audio-Reasoner	**0.207**

4.4.1 Key Findings

Audio-based models performed better than the feature-based LLaMA 3 approach on spontaneous Spanish speech.

Audio-Reasoner achieved the best balanced accuracy at 63.90%, the highest AUROC at 0.676 and the lowest Brier score at 0.207.

Qwen2-Audio achieved 100% specificity, meaning that no healthy controls were incorrectly classified as PD. However, its sensitivity was only 26.09%, meaning that most PD participants were missed.

4.4.2 Interpretation

Spontaneous speech contains more complex temporal and prosodic information than highly constrained speech tasks.

Raw audio models may therefore have an advantage because they can potentially capture:

- temporal rhythm;
- pauses;
- pitch variation;
- loudness variation;

- articulation dynamics;
- speech rate; and
- interactions between multiple acoustic characteristics.

Handcrafted feature pipelines may compress some of these characteristics into summary statistics and therefore lose certain temporal patterns.

5 Cross-Dataset Analysis and Patterns

5.1 Modality Effects Summary

Table 14: Summary of modality effects across the four datasets.

Dataset	**Best Modality**	**Performance Gap**	**Key Pattern**
BenSParX	**Feature-based**	+33.33% B.Acc	Feature-based approach is substantially more stable for low-resource Bengali speech.
MDVR-KCL	**Audio-based**	+18.75% B.Acc	Audio reasoning performs better on controlled English read speech.
IPVS	Mixed	Small difference	Dataset-dependent behavior with strong threshold effects.
NeuroVoz	**Audio-based**	+11.32% B.Acc	Raw audio models perform better on spontaneous speech.

The results clearly demonstrate that no single input modality dominates across all datasets.

5.2 Consistent Patterns

5.2.1 Feature-Based Approaches

Several broad characteristics emerge for feature-based zero-shot inference.

- More stable behavior in low-resource settings.

- Strong AUROC in several datasets.
- Better calibration in the Bengali dataset.
- Potentially greater interpretability because individual features have known acoustic meaning.
- Vulnerability to threshold-related sensitivity-specificity trade-offs.

The Bengali results are particularly notable because the feature-based LLM achieved substantially better discrimination than all audio models.

5.2.2 Audio-Based Approaches

Audio-based models demonstrated:

- Dataset-dependent improvements.
- Higher balanced accuracy in MDVR-KCL and NeuroVoz.
- Greater sensitivity to dataset characteristics.
- Greater variability in calibration.
- Potential advantages for spontaneous speech.

The results suggest that raw audio provides access to information that may be lost during handcrafted feature aggregation. However, access to more information does not guarantee robust zero-shot classification.

5.3 Model-Specific Patterns

Table 15: Observed model-specific behavior.

Model	**Observed Pattern**
LLaMA 3	Strong discrimination in several datasets but substantial threshold-dependent sensitivity-specificity trade-offs.
Audio-Reasoner	Strong balanced accuracy on multiple datasets and improved performance in some raw-audio settings.
Pengi	Consistently weak zero-shot generalization across the evaluated datasets.
Qwen2-Audio	Highly variable performance with a tendency toward strong specificity in several settings.

5.4 Calibration versus Discrimination

One of the most important findings is that improvements in balanced accuracy from raw audio input are not consistently accompanied by improvements in calibration.

This distinction can be expressed conceptually as:

$$\text{Good discrimination} \neq \text{Good calibration.} \tag{1}$$

A model may correctly rank PD and healthy subjects but still assign poorly calibrated probabilities.

The results therefore demonstrate three distinct properties:

1. **Discrimination:** Can the model distinguish PD from healthy speech?
2. **Classification:** Does the selected decision threshold produce correct labels?
3. **Calibration:** Do predicted probabilities correspond to actual outcome frequencies?

For clinical decision-making, all three properties are important.

6 Critical Analysis

6.1 Strengths

The unified zero-shot evaluation has several important strengths.

1. **Systematic Comparison:**

   It provides a unified comparison of input modalities for zero-shot PD detection.

2. **Multilingual Evaluation:**

   Four languages are included: Bengali, English, Italian and Spanish.

3. **Multiple Speech Tasks:**

   The evaluation includes conversation, reading, text-dependent utterances and spontaneous speech.

4. **Statistical Rigor:**

   The evaluation uses bootstrap confidence intervals and multiple complementary metrics.

5. **Modality Effects Isolated:**

   The same handcrafted feature pipeline is used across datasets, reducing the influence of dataset-specific feature engineering.

6. **Clinical Relevance:**

   Brier score is included to assess calibration in addition to discrimination and classification performance.

7. **Practical Insights:**

   The experiments reveal which modality may be more effective under different language and speech-task conditions.

6.2 Limitations

Despite the methodological strengths, several limitations remain.

1. **Strict Zero-Shot Setting:**

   No fine-tuning or few-shot learning was investigated. Consequently, the results may underestimate the ultimate potential of the models.

2. **Modest Dataset Sizes:**

   The datasets contain relatively small numbers of subjects, which can produce substantial uncertainty even when bootstrap confidence intervals are used.

3. **Fixed Feature Pipeline:**

   The handcrafted features were standardized rather than individually optimized for each corpus. This is useful for isolating modality effects but may limit the maximum performance of feature-based methods.

4. **Limited Model Selection:**

   Only four models were evaluated: one LLM, two LALMs and one LARM.

5. **No Task-Specific Adaptation:**

   Fine-tuning or domain adaptation could potentially improve audio-model performance.

6. **Limited Language Coverage:**

   Four languages do not represent the linguistic diversity of the global population.

7. **No Clinical Validation:**

   Prospective clinical validation is required before these systems can be considered for deployment.

6.3 Ethical and Clinical Considerations

Zero-shot foundation models raise several ethical and clinical concerns.

The models should not be used for clinical diagnosis without rigorous prospective validation and appropriate regulatory and clinical oversight.

6.3.1 Reliability

The large performance variation across datasets demonstrates that zero-shot models cannot currently be assumed to generalize reliably to unseen populations.

6.3.2 Bias

Extreme sensitivity-specificity trade-offs may produce systematic errors. For example, a model with very high sensitivity but very low specificity could generate a large number of false-positive PD predictions.

Conversely, a model with high specificity but low sensitivity may fail to identify a substantial number of individuals with PD.

6.3.3 Transparency

Large language and audio models are substantially less transparent than traditional feature-based models. Understanding which acoustic characteristics drive a prediction is particularly important in a medical setting.

6.3.4 Clinical Validation

Prospective validation using independent clinical populations is essential before deployment. Laboratory performance alone cannot establish clinical utility.

7 Broader Context and Related Work

7.1 Foundation Models in Parkinson's Disease Research

The emergence of foundation models has also influenced supervised and transfer-learning approaches to PD detection.

Table 16: Selected foundation-model studies related to PD detection.

Study	**Approach**	**Key Finding**
Favaro et al. (2023)	Interpretable acoustic features versus DNN embeddings	DNN embeddings can outperform interpretable acoustic features in multilingual scenarios.
La Quatra et al. (2024)	Foundation models for PD detection	WavLM Base achieved an AUC of approximately 0.82 on PC-GITA.
Purohit et al. (2025)	Foundation-model layer selection	Whisper embeddings combined with an adapter achieved approximately 85% accuracy.
Dao et al. (2025)	Ensemble of Wav2Vec 2.0, Whisper and SeamlessM4T	Reported AUROC of 91.35% on the ICEBERG dataset.

These studies demonstrate that pretrained speech representations can be highly useful when combined with supervised adaptation. However, supervised or fine-tuned systems should not be directly equated with zero-shot systems because the amount of task-specific information available to the model differs substantially.

7.2 Zero-Shot Applications in Healthcare

Zero-shot foundation models are increasingly being explored across healthcare applications.

Table 17: Examples of zero-shot foundation-model applications in healthcare.

Domain	Application	Current Status
Cognitive Impairment	Speech-based detection	Promising but early-stage.
Depression	Audio-language-model zero-shot classification	Under active investigation.
Mental Health	Tone and prosody analysis	Active research area.
Clinical Decision Support	LLM-based diagnostic reasoning	Potentially useful but associated with major ethical and safety concerns.

The broader healthcare literature suggests that foundation models have substantial potential, but their clinical reliability and safety remain open questions.

7.3 Key Differences from Supervised Approaches

Table 18: Supervised learning versus zero-shot foundation models.

Aspect	Supervised ML/DL	Zero-Shot Foundation Models
Training Data	Task-specific labeled data	General-domain pretraining data.
Adaptation	Feature engineering and parameter optimization	Prompt engineering.
Language	Often requires language-specific training data	Potential for cross-lingual transfer.
Calibration	Can be optimized using validation data	Often less stable without task-specific calibration.
Interpretability	Feature-based models can be relatively interpretable	Large foundation models are generally less transparent.
Generalization	Often optimized for distributions similar to training data	Potentially broader but highly inconsistent.

The most important distinction is that supervised systems are explicitly optimized for the target disease classification problem, whereas zero-shot systems rely almost entirely on representations acquired during general pretraining.

8 Implications and Future Directions

8.1 Practical Implications

8.1.1 Model Selection Depends on Use Case

The experimental results suggest that model selection should depend on the target application.

Table 19: Potential modality choices under different deployment scenarios.

Scenario	**Potentially Preferable Approach**
Low-resource languages	Feature-based approaches may provide greater stability.
Spontaneous speech	Audio-based models may benefit from direct access to temporal information.
Controlled speech	Performance is strongly dataset-dependent.
Clinical deployment	Neither modality should be deployed without calibration and prospective validation.

8.1.2 Calibration Matters

High accuracy or balanced accuracy alone is insufficient for clinical applications.

The inclusion of the Brier score demonstrates why probabilistic calibration should be evaluated alongside classification performance.

Future clinical systems should therefore report:

- balanced accuracy;
- AUROC;
- sensitivity;
- specificity;
- precision;
- F1-score; and
- Brier score or other appropriate calibration measures.

8.1.3 Zero-Shot Learning Is Not Sufficient Alone

The results demonstrate that zero-shot foundation models are promising but currently unreliable for clinical PD screening.

A practical development pathway may therefore involve:

Zero-Shot → Few-Shot → Domain Adaptation → Prospective Validation → Clinical Deployment.

Each stage introduces additional information and validation before the model is considered suitable for real-world use.

8.2 Future Research Directions

8.2.1 Few-Shot and In-Context Learning

Future studies should investigate whether providing a small number of labeled examples can substantially improve zero-shot performance.

Few-shot learning may provide an intermediate solution between completely zero-shot inference and full supervised fine-tuning.

8.2.2 Domain Adaptation

Foundation models trained on general speech may benefit from adaptation to pathological speech.

Potential approaches include:

- parameter-efficient fine-tuning;
- adapters;
- contrastive learning;
- domain-specific pretraining; and
- language-specific adaptation.

8.2.3 Prospective Validation

Future studies should evaluate models on independent clinical cohorts collected prospectively.

This is necessary to determine whether reported performance remains stable under realistic clinical conditions.

8.2.4 Interpretability

Interpretability methods should be developed to determine which speech characteristics influence model predictions.

Potential methods include:

- acoustic feature attribution;
- attention analysis;
- perturbation-based analysis;
- saliency methods; and
- comparison between model explanations and clinically established biomarkers.

8.2.5 Expanded Language Coverage

Future datasets should include a much broader range of languages and language families.

Particular attention should be given to:

- low-resource languages;
- underrepresented geographic regions;
- multilingual speakers; and
- culturally diverse speech patterns.

8.2.6 Spontaneous Speech

Naturalistic speech is particularly important because real clinical deployment is unlikely to be restricted to sustained vowels or carefully controlled utterances.

Spontaneous speech can provide information about:

- prosody;
- speech rhythm;
- pauses;
- articulation;
- lexical organization;
- speech rate; and
- conversational dynamics.

8.2.7 Longitudinal Studies

Longitudinal studies could determine whether speech-based foundation models can monitor disease progression rather than merely classify PD versus healthy controls.

Such studies could investigate whether changes in model-derived speech representations correlate with:

- disease progression;
- clinical severity;
- treatment response; and
- changes in motor and non-motor symptoms.

8.3 Recommendations

Table 20: Recommendations for different stakeholders.

Stakeholder	**Recommendation**
Researchers	Report calibration metrics alongside accuracy, investigate few-shot learning and standardize evaluation protocols.
Clinicians	Interpret zero-shot predictions cautiously and require prospective clinical validation.
Regulators	Require rigorous independent validation and appropriate safety evaluation before clinical approval.
Developers	Focus on calibration, explainability, robustness and demographic fairness rather than accuracy alone.

9 Conclusion

This study provides a systematic comparison of input modalities for zero-shot Parkinson's disease detection from speech using large language and audio models.

Several important conclusions can be drawn.

1. **Modality Effects Are Dataset-Dependent:**

   Feature-based approaches provide more stable zero-shot behavior in the Bengali low-resource setting, whereas audio-based approaches demonstrate advantages on some English and Spanish datasets.

2. **Calibration Is Critical:**

   Higher balanced accuracy does not necessarily correspond to better-calibrated predictions. Therefore, the Brier score and other calibration measures should receive greater attention in clinical speech research.

3. **Zero-Shot Learning Alone Is Insufficient:**

   Current zero-shot capabilities do not provide sufficiently consistent performance to justify direct clinical deployment.

4. **Language Matters:**

   Performance varies substantially between languages. Low-resource languages may present particularly difficult conditions for large audio models.

5. **No Universal Best Model Exists:**

   Different models perform best under different combinations of language, speech task and recording condition.

The results reinforce the idea that the choice between handcrafted acoustic features and raw audio should not be viewed as a universally settled question.

Instead, the optimal representation depends on the interaction between:

$$\boxed{\text{Language} + \text{Speech Task} + \text{Recording Condition} + \text{Model} + \text{Input Representation}}$$

A particularly important finding is that a general-purpose LLM operating on a relatively simple acoustic representation can outperform much larger audio-capable foundation models in a low-resource language. This suggests that clinically meaningful acoustic

biomarkers may retain considerable value even in the era of large multimodal foundation models.

Conversely, the improved performance of audio-based reasoning models on spontaneous speech indicates that direct access to raw acoustic information may become increasingly valuable when the speech task contains complex temporal and prosodic information.

Ultimately, zero-shot foundation models represent a promising new direction for speech-based PD detection, but their current performance is not sufficiently reliable for clinical deployment.

Future work should focus on:

- few-shot and in-context learning;
- domain-specific adaptation;
- calibration and uncertainty estimation;
- interpretable foundation models;
- multilingual and low-resource speech;
- spontaneous and naturalistic speech;
- longitudinal disease monitoring; and
- prospective clinical validation.

The long-term goal should therefore not simply be to achieve higher classification accuracy. Instead, research should move toward systems that are **robust, calibrated, interpretable, fair, reproducible and clinically validated**.

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